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Micol Mieli, Malin Zillinger & Jan-Henrik Nilsson

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RESEARCH ARTICLE

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Phygital time geography, or: what about technology in tourists' space-time behaviour?

Micol Mieli , Malin Zillinger and Jan-Henrik Nilsson

Department of Service Studies, Lund University, Helsingborg, Sweden

ABSTRACT

The paper argues for the renewed relevance of time geography in tourism in light of the use of mobile technologies and ubiquitous connectivity. The paper proposes the concept of phygitality to understand how digital technologies are used in physical space, and how the interaction between the physical and the digital reconfigures tourists' projects, paths, bundles, and constraints. The theoretical contribution builds on fifteen semi-structured interviews. The analysis shows that capability, coupling, and authority constraints are altered and mediated by digital devices. In the phygital time-space, tourists orient themselves in physical spaces, influenced by digital information; they create phygital paths and move between stations that result from the overlaying of digital information onto the physical space. Tourists' goal-oriented mobility results in phygital projects, where logics of efficiency and optimization reduce the liminality of the tourist experience. Tourists' bundles are created within and outside the physical vacation prism through digital communication.

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Introduction

Tourists' spatial behaviour has changed radically due to the use of smartphones (Ash et al., 2018; Caldeira & Kastenholz, 2020; Kang, 2016; Wang et al., 2016). This is interesting for tourism geographies, because while mobility in space and time is the defining element of tourism and travel, mobile technologies that offer ubiquitous connectivity have altered the way people experience space (de Souza e Silva, 2006; Frith, 2012; Gaggioli, 2017). Human mobility, in tourism and elsewhere, can be understood from many different perspectives. With its focus on individuals' movement and interaction in time-space, we argue that time geography (see Ellegård, 2018; Hägerstrand, 1970, 1973; Hallin, 1988) offers useful tools to understand how time and space are experienced by tourists in a technologically mediated experience. Therefore, this paper claims that once again time geography is significant to understand tourism and its conceptualisation should be adapted to the digital context, with particular

CONTACT Micol Mieli  micol.mieli@ses.lu.se  Department of Service Studies, Lund University, Helsingborg, Sweden

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attention to how physical and digital dimensions are intertwined. Therefore, this paper conceptualizes time geography in tourism with an adaption to digitalisation.

Digitalisation enables tourists to engage in different contexts simultaneously, for example when they are driving through the destination and simultaneously take part in an online meeting at work (Liu et al., 2022). In doing so, the digital and the physical do not only exist side by side, but they are enmeshed, overlaid, and merged. This is thought-provoking, as the digital and the physical have long been considered as the opposite of one another. We address this conceptual challenge by using the concept of phygitality, a playful combination of the words physical and digital that indicates how the physical world is turned into a hybrid, multi-dimensional context through the use of smartphones (Andrade & Dias, 2020; Mieli, 2022a; Liu et al., 2022; Lo Turco & Giovannini, 2020; Nofal et al., 2017; Zurlo et al., 2018). Given the ability of smartphones to instantly connect people in different places, and to promptly transfer them to different digital settings, tourists can increasingly interact simultaneously with people, places, and contexts that are located somewhere else. They do so even if each individual body is spatially fixed to one place only (Dickinson et al., 2014; Liu et al., 2022). For example, tourists can be camping with their family, and spend time with friends on the other side of the world through postings on social media. Hence, tourists are not either in a physical or in a digital world, but in an intersection of them. Digital communication, moreover, depends on material means, such as computers and mobile phones, which complement the traditional material means (e.g. transport) considered in time geography (Couclelis, 2009).

This article departs from the notion that smartphones have a profound impact on both physical and the perceived space in which tourism takes place. Smartphones transform the physical nature of tourism through their constant connectivity, portability, and location-based services. They also alter potentials and constraints of tourists' behaviour in time-space, which previously has been understood as purely physical (Vaez et al., 2020). We call this space, that tourists travel through and in which they are digitally referring to other places at the same time, a phygital space. This phygital space changes how we can understand humans' mobility, connectivity, and mooring in time and space. Research has shown that smartphones are not only uncoupling individuals and places, but actually recoupling social interaction (Thulin et al., 2020, p. 173). While uncoupling here implies that online contacts are enabled anywhere, at any time, recoupling means that for many social contacts, individuals still need to be online concurrently, for example in chats. Moreover, information and communication technologies are loosening the link between activity, place, and time (Couclelis, 2009). Therefore, we discuss tourists' phygital behaviour from a time-spatial perspective.

The aim of the article is to conceptualize phygital time geography, adapted to technologically mediated tourists who travel with their smartphones (cf. Mieli, 2022a). To address this aim, we analyse how tourists with smartphones integrate digital and physical worlds in their time-spatial behaviour. In this, we build on the concepts of paths, bundles, and constraints to operationalize the rather abstract time geography model. The concepts are related; in real life, the development of projects and paths, and the formation of bundles are dependent on various forms of constraints as 'paths become captured within a net of constraints' (Hägerstrand, 1970, p. 11). The study employs an explorative approach to data collection and has no aims to be generalisable. Due to

the ambition to contribute to the discussion on adapting time geography to digitalisation in tourism, significant weight is given to conceptual reflections.

The theoretical section begins with a presentation of time geography. Thereafter comes a literature review of smartphones and tourists' spatial behaviour. The methodology shortly describes the use of semi-structured interviews for data collection. In the analysis, empirical data are presented and discussed in relation to time-spatial perspectives. A discussion of the empirical results in connection to the theoretical framework concludes the paper.

Time geography

Time geography offers a perspective that helps us describe and explain human movements in the physical world. It takes into account the conditions in which mobility takes place. Time geography emphasizes the importance of time and space for social science analysis, particularly when it comes to revealing relations between people, places, and activities. It departs from a physical realistic view of the world, however with considerable room for human agency (Ellegård, 2018; Hallin, 1988). Note that it is not an explanatory theory, but an 'ontological contribution preceding formation of a theory' (Hägerstrand, 1985, p. 195). It rests on four basic principles: (i) time and space are two inseparable dimensions; (ii) all individuals and objects have limited time spans; (iii) space is a limited resource; and (iv) all individuals and objects are essentially in-divisible (Hallin, 1988). In phygital space these principles can be questioned, because people can jump digitally between places.

Time geography is associated with the time-geographic model, in which '[h]uman actors [...] can be observed physically and portrayed in a time-geographic notation as they move around' (Lenntorp, 1999, p. 155). It becomes a 'means to keep track of both the spatial and the temporal dimensions simultaneously' (Lenntorp, 1999, p. 156). In their daily lives, people perform various *projects* that involve mobility, for example travelling to work or to a destination, or visiting an attraction. In performing these projects, they move in sequences along *paths* between *stations*. People can for instance walk between their hotel and a restaurant. The path concept visualizes individuals' time-space movements, where stations have both time and space coordinates (Ellegård, 2018). As people move between stations, their time-spatial movements look like *prisms*. The concept of prism is used to indicate both past, present, and future locations in time-space (Ellegård, 2018). Depending on the distance people move, prisms are decreasing or increasing in size (cf. Figure 1). The speed of movement affects the graphic angle of the paths. Given that a tourist trip is defined in space and time, we can say that tourists have a 'vacation prism' within which they move and form bundles with other people, through physical and/or digital interaction. Paths can be regarded as flows of events; however, a path cannot cross the now-line (Ellegård, 2018). When individuals encounter and come together in one place, their paths gather in *bundles* (Hägerstrand, 1970, 1973). To Hägerstrand, encounters were essentially physical, about getting *in touch*, and based on the assumption that individuals are indivisible (cf. Hallin, 1988). Therefore, individuals cannot be in more than one place at a time. Digitalisation challenges such assumptions.

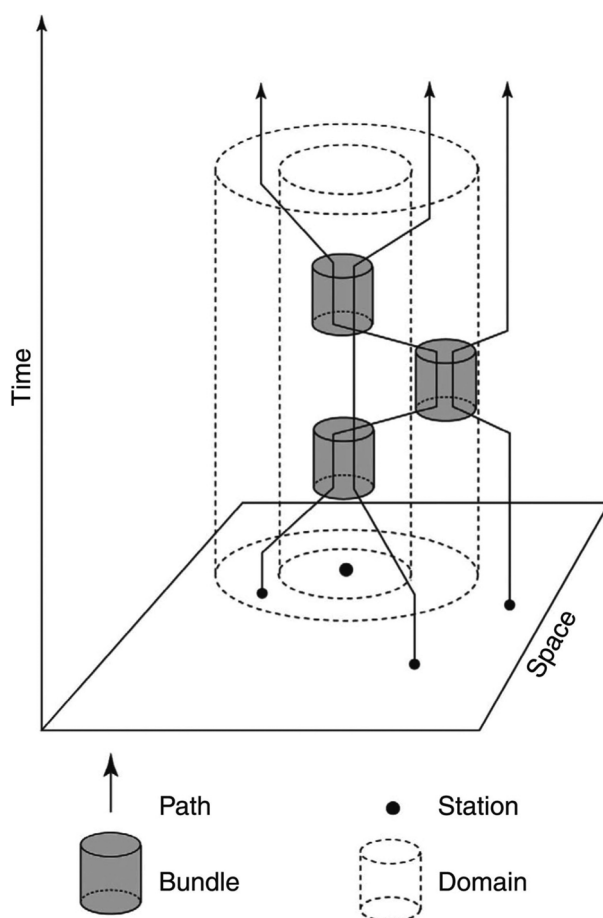


Figure 1. Hägerstrand's time-geographical diagram. In Shoval (2012, p. 176).

People's movements in time-space are limited by three types of interrelated *constraints*. *Capability* constraints 'are those which limit the activities of the individual because of his biological construction and/or the tools he can command' (Hägerstrand, 1970, p. 12). They are primarily embodied, based on individuals' physical and mental capacity but they also relate to resources, such as tools and means of communication. In relation to mobile phones, people are constrained by how long their phones can last without being recharged (Schwanen & Kwan, 2008). *Coupling* constraints refer to 'where, when and for how long a person needs to join other individuals, tools and materials in order to produce, consume and transact' (Hägerstrand, 1970, p. 14). The capacity for people to form bundles is highly related to coupling constraints. As actions are coordinated between travel parties, group composition has been interpreted as a coupling constraint in previous tourism research (Shoval, 2012). In a similar vein, Zhang and Zhang (2022, p. 2) argue that digital technologies enable people to come together in 'inconsistent space-time', meaning individuals are socially co-present in different places. This approach resembles Thulin and Vilhelmson (2019) understanding of recoupling.

The third type, *authority* constraints, is related to time-geographic aspects of power, for example over the access to certain places. For Hägerstrand (1970, p. 16) authority constraints fill the world with *domains* in time-space that are under the control of individuals, groups, or authorities. For example, it may be forbidden to take photos or to answer phone calls in churches or museums (Schwanen & Kwan, 2008). Authority constraints are also affected by social barriers and norms, conditions that indirectly restrict access to various places (Grinberger et al., 2014).

Although time-geography was originally about everyday life, it has been used in tourism studies as well. Through the decades, tourism researchers have fruitfully come to use time geography to understand tourism mobility (Zillinger, 2007; Petterson & Zillinger, 2011; Grinberger et al., 2014; Hall, 2005; Kang, 2016; Qviström et al., 2020; Shoval, 2012; Xiao-Ting & Bi-Hu, 2012). However, most of these studies were done before the appearance of smartphones.

Research on time geography in the digital age has found that new constraints are created by the pervasive use of technologies (e.g. Schwanen & Kwan, 2008; Thulin & Vilhelmson, 2019). Thulin et al. (2020) explore the consequences of smartphone use on young people's everyday constraints and conclude that it leads to a reordering of time, space, and place. Smartphone use leads to decoupling of people and places and to a more flexible use of space, but it brings along other constraints connected to changing rhythms of social interaction. Thulin and Vilhelmson (2019) analyse how background and foreground activities are conducted simultaneously and in parallel, in bundles of online and off-line activities. They call for a theoretical development of digital geography, which we aim to address. We therefore use time-geographic concepts to operationalize our account of tourists' digital behaviour in phygital space.

Towards a phygital time geography: smartphones and space

The term phygital is used to indicate the ways in which physical and digital elements blend to create something qualitatively new (Andrade & Dias, 2020; Mieli, 2022b; Zurlo et al., 2018). The digital transformations of almost all contexts of life blur the distinction between the physical and the digital, and lead to a living space that is 'digitally enriched' (Gaggioli, 2017, p. 774). This results in a new concept of space because experiences in one place are shaped by the digital presence in another. One can say that experiences in physical space are supplemented by an additional digital layer. While the term phygital often appears in marketing and heritage studies (Belghiti et al., 2017; Lo Turco & Giovannini, 2020; Nofal et al., 2017), it has only recently been used in tourism (Ballina et al., 2019; Gretzel et al., 2019; Neuburger et al., 2018). Except for Gaggioli's (2017) short piece on phygital spaces, it seems almost absent in geographic scholarship.

One needs to understand physical and digital space not as separate but as one and the same. Tourists are on the phone *while* they are at the destination, and digitally at work *while* lying by the pool. Through smartphone mediation, several contexts blend into the vacation context, creating hybrid spaces that can be defined phygitaly (Mieli, 2022a). In this way, tourists' relations to time, space, and place are reordered when they take part in multiple places at the same time. It is argued that this phygital

situation influences individuals' possible projects in timespace. Hence, phygital tourist behaviour revises the constraints that would otherwise restrain tourists' spatial mobility. This paper studies whether and how this is done.

Smartphones influence how tourists move in time and space, as well as how tourists connect to their peers at home (Anaya & Lehto, 2020). Several authors (e.g. Couclelis, 2009; Zhang & Zhang, 2022) observe how tourists can have a feeling of being simultaneously at home and away, through their mobile phones. In fact, information and communication technologies constitute a link between people and places. While this may make physical distance irrelevant, psychological and emotional consequences are more complex, as people feel both far and near (White & White, 2007). Conti and Heldt Cassel (2020) illustrated how social media use affects tourist spatialities and space-making, especially by influencing the role of liminality in nature-based experiences. The digital elasticity of mobile phones has the power to displace the concept of liminality, as it mitigates the effects of travelling to a new place (Pearce, 2011). This possibly removes the sense of magic and escape associated with travel (Benckendorff et al., 2019). In conclusion, mobile technologies transform tourists' 'experience of space by enfolding remote contexts inside the present context' (de Souza e Silva, 2006, p. 262). This phenomenon of feeling co-present occurs when tourists' attention is divided between their physical and digital contexts (Lamsfus et al., 2015; Sheller & Urry, 2006).

Smartphone use also causes a spillover effect of everyday functions and activities into travel, including communication with distant people, or mundane activities, such as online banking and work-related email conversations (Mieli, 2022c; Wang et al., 2016). This enables travel that otherwise had been impossible to conduct because of the necessity for co-presence at home or in the workplace (Tan & Chen, 2021). This is what is so peculiar with mobile technologies: they offer the possibility to move through space, at the same time as one is interacting with others in remote places.

Through their ability to provide contextualized, personalized spatial information to tourists, smartphones can transform tourists' interactions with places, activities, and people. These hybrid spaces can change the way physical space is negotiated and understood (Frith, 2012). For example, Duignan et al. (2018) showed how digital spaces complemented physical dittos to overcome the dichotomy between core and peripheral locality at food festivals. Several studies have focused on wayfinding strategies with different navigation tools (Chang, 2013; Ishikawa & Montello, 2006; Vaez et al., 2020). Tourists who use GPS recognize less spatial information and use fewer strategies than those who only rely on analogue sources. They prefer to follow the route suggested by the device rather than exploring on their own (Vaez et al., 2020). However, such rational logic is usually believed to be foreign to the tourist experience and to belong to everyday life only (Zhang & Zhang, 2022).

Chen et al. (2021) showed that digital maps influence tourists' use of landmarks for wayfinding. In addition, they showed that map choice in combination with spatial familiarity with the place influences people's affinity to make turns along their way. Tourist information that identifies an attractive site, particularly in digital maps, hold considerable power over tourists' choice of routes and of 'what ought to be seen' (Koshar, 1998). Hence, such maps become tools for producing specific

types of behaviour following indications on the map (Baum & Ribak, 2021; Vaez et al., 2020). Moreover, Lamsfus et al. (2015) acknowledge the ability afforded by smartphones to micro-coordinate activities with other people both outside and inside the travel party.

These examples show that empirical studies on digitalisation and spatial behaviour exist, but to the best of the authors' knowledge, the conceptual understanding thereof is thin, as also stated by Thulin and Vilhelmson (2019). The referenced literature shows how smartphones influence tourist behaviour in space. We argue that their use should also influence the ontological understanding of space along a tour. To enhance our understanding of these relations, we use the concept of phygitality to further explore tourists' time-space behaviour in our empirical study.

Methodology

The empirical data consists of fifteen semi-structured interviews performed after the tourists' journeys. Data was collected between May 2019 and March 2020 within a larger study on tourist behaviour and smartphones. In that study, the influence of smartphones on tourists' spatial perceptions and actual behaviours was noted; hence, this paper. For detailed information on data collection, see Mieli (2022c). Participants were born in the 1980s and 1990s, roughly falling within the so-called generation Y or millennials. This demographic group uses the internet more than any other during their travels (Kang et al., 2021; Kim et al., 2015). Moreover, this generation has experienced travel both with and without smartphones and is therefore believed to be aware of digital impacts on the experience (Mieli, 2023; Bakker, 2019). They are comfortable with the use of smartphones and are used to being constantly connected to the internet but were born before this became the norm (Schulz et al., 2019). All participant names were anonymized using nicknames.

Interview questions departed from smartphone use for information related purposes, but expanded to include general use of smartphone. Interviews were conducted by one of the authors using an interview guide, which was updated throughout the research process to account for emerging themes, according to the principle of constant comparison (Bryant & Charmaz, 2007). All interviews were recorded and transcribed.

The thematic analysis was conducted in three steps: a first round of *in-vivo* coding was carried out by one of the authors and then further rounds of coding and analysis were conducted in collaboration with the other authors to ensure inter-coder reliability (cf. O'Connor & Joffe, 2020). The first round of *in-vivo* coding resulted in 84 codes, which were then elaborated into a codebook consisting of 42 codes and sub-codes, grouped into nine broad themes: connectivity, basic materialities, dis/connect, spatial orientation, cartographical aspects, sense of location, placemaking, occasioning use/trigger, work. These guided the analysis in the second step of thematization, and were further elaborated into fewer themes, informed by theory, such as 'physical capabilities of digital devices', 'effective use of time', and 'separability of time and space'. The last step, theory-building, consisted of the conceptual work of connecting the analysis of the empirical material to the theory of time geography, from which the three main analytical themes emerged: (1) constraints; (2) paths; and (3) bundles.

Analysis

In this section, we discuss technologically mediated tourists’ time-spatial behaviour and include quotes from the interviews along with the argumentation. The analysis is divided into three main themes: constraints, paths, and bundles. The themes and related quotes are summarized in tables and further explained in the following paragraphs.

Constraints

Besides their embodied and transport-related capabilities, modern tourists need to consider the capabilities of their devices when travelling. We have identified three main determinants of smartphone capabilities: hardware, software, and infrastructure. The quotes included in Table 1 are only a few examples of how several participants acknowledged this.

Data showed that battery duration, internet connectivity, device performance, and operation of Global Positioning System (GPS) were important factors that tourists consider when travelling with their smartphones. Tourists can extend such capabilities, for example by extending battery life through additional tools (Amira 1). The speed of the internet connection and the performance of the device and its applications, such as digital maps, are also important (Anastasia 2). In this context, new capability constraints emerge from hardware, software, and infrastructural capabilities connected to smartphones. Using online maps not only allows tourists to know where they are in space but also to be efficient in their use of time and space. Also, they can keep record of distances. Moreover, although people can work more easily during their trips, they are still constrained by basic requirements: they need electricity and a physical place to work from. As noted by Schwanen and Kwan (2008), material technologies need to be placed in physical space-times. This is where the concept of cyberspace falls short: merely considering the digital space is not enough to understand the physical and digital constraints on tourists’ activities.

Internet connections can also be limited by infrastructural factors. Primarily, the so-called ‘grid’ has to be in place through antennas, poles, and cables that can convey signals across space (Fritz 3). Infrastructure also imposes legal or organisational constraints to conducting certain activities in specific spaces that are connected with domains of authority in phygital space. An obvious example is that of EU regulations on roaming fees and internet costs while travelling (Zillinger et al., 2018). Private contracts with phone providers, and Wi-Fi connections in cafés and hotels, also

Table 1. Participants’ quotes on constraints.

Theme	Sub-theme	Quotes
Constraints	Hardware	1. Amira ‘Actually, I have two phones and a power bank all the time in case I run out of battery’
	Software	2. Anastasia ‘this time we were using google maps and we were in a hurry and we didn’t know where to go exactly, (...) we used google but we realized that we were running out of time so we had to hurry but the phone was not so quick so we ended up doing several circles’
	Infrastructure	3. Fritz ‘It was amazing in Scotland. I was in the most remote areas of Scotland on some small islands, with no people, no civilization, no nothing, but there was always 4G and it worked perfectly everywhere.’

constrain the capabilities of phones and the domains within which tourists use their phones. Authority constraints also arise from legal or organisational constraints to working abroad, such as matters of insurance, or company policy.

Paths

The most important influence of smartphone use is related to individuals' spatial behaviour and perception. Tourists described how digital maps changed the way they perceive the places they visited and the way they perceive the time dimension of their movements (Table 2). The analysis of tourists' paths yielded three additional sub-themes: physical orientation through digital information; phygital stations, goal-oriented mobility projects.

Physical orientation through digital information. We found that individuals changed their way of orientation through digital maps, including the ability to locate oneself in space (Sofia 4, Ines 5). What we see is a perceived inability to find a spot, for instance, an attraction, if the smartphone does not operate correctly (Hazel 7). Kaisa (6) explains how she was able to localize a place with the help of paper maps, but that she has a far lower level of spatial understanding when using digital maps on her smartphone. In other words: tourists depend on digital maps, and are less equipped to find a spot either by locating landmarks or by leaning on a sense of direction. Just like Ishikawa and Montello (2006) and Vaez et al. (2020), we found that tourists with digital devices are less attentive to their geographical surroundings along their paths. As a consequence, tourists may value mobility through new and interesting surroundings more when neglecting

Table 2. Participants' quotes on paths.

Theme	Sub-theme	Quotes
Paths	Physical orientation through digital information	1. Sofia 'I think London is so big that occasionally I went on [the online map] to see where we were, to locate myself... to zoom out and oh now we are in the West'
		2. Ines 'I could by no means tell you in which direction something was. Because I was so focused looking on the [digital] map that I forgot to actually look around'
		3. Kaisa 'When I was younger we always used to take paper maps from cities when we travelled and I would always know after one or two days where everything was because I would remember the map that well and nowadays I can't do that anymore, I don't know if it's because I don't use enough time for that or I'm just stupider (...) also because if you turn around you can see the map turning around'
		4. Hazel 'The most important thing (...)It's directions (...)because the city is so dense and there's so many roads so sometimes it's a bit confusing so a lot of time was spent looking for directions' (...)If i didn't have a smartphone i don't know if we would have survived in Vietnam'
	Paths and stations	5. Ines 'There was one day in Berlin that I saw a food market and I told [my boyfriend] ahh let's go see it, and he's like yeah but it doesn't appear here [on Google maps], maybe there aren't that many things to see'
		6. Gabriela 'I was only limited to seeing things that Google maps showed me. I'm pretty sure that there were so many more things that I missed out on just because I was only using this platform'
	Goal-oriented mobility: projects	7. Olof 'Google maps makes you navigate the fastest possible way, but it doesn't matter so much if you're on vacation. Maybe driving around a bit could be really nice [...] instead of Google maps-ing'
		8. Alma 'We decided with the distances, like we will go to this one and this one but we checked with the phone. We wouldn't drive around too much or get lost, we needed to know places around that we want to see that day'

smartphones for a while. Although tourists usually find their way without phones, they feel less secure about their ability to do so (Ines 5, Kaisa 6).

Stations along the path. In time geography, the concept of paths is used to trace and visualize time-space movements (Ellegård, 2018). As they travel along paths, tourists reach their chosen stations. Stations are places where people spend time and get together—they are attractive for some reason, so individuals may need to gather information about them. In the context of tourist experiences, stations can be called attractions, sites, or destinations. Interactive, online maps that provide location-based services (LBS) can be seen as authoritative sources of information about what is and what is not a place to visit. This was the case of Ines (8) (in Table 2), who argued with her travel companion about whether the food market was a place to visit simply because it was not shown on Google Maps. Ines and her boyfriend ended up visiting the food market anyway and found it a rewarding experience. Therefore, while these maps effectively instruct tourists on ‘what ought to be seen’ (Koshar, 1998), as Gabriela (9) clearly acknowledged, tourists still have agency in negotiating the digital and physical dimensions of spatial information. Along paths in phygital space, tourists therefore move between physical and digital space, where location-based services can place them in physical space and add layers of real-time spatialized information onto that space (Mieli, 2022a; Frith, 2012).

Goal-oriented mobility: phygital projects. In their journeys along paths and between stations, tourists plan and carry out their projects. In line with previous tourist studies on wayfinding (Chang, 2013; Ishikawa & Montello, 2006; Vaez et al., 2020), our data show that using online maps not only allow tourists to locate themselves in space but also to achieve an efficient use of time and space in organising and conducting their projects. Olof (11) and Alma (12), for example, note how their physical mobility project is influenced by digital tools, which allow them to rationalize and optimize their use of time as they move in space. In fact, time is used efficiently because the fastest or optimal route between two stations can be calculated by the device through location-based services.

Reaching the destination is the primary goal, and this entails that less serendipitous experiences may happen along the way (cf. Mieli, 2023). When tourists follow the nearest way, this also means that their walking prism becomes slim. Phones promote efficiency since they do not show any unplanned detours; the possibility of getting lost and wandering about at the destination can be almost eliminated. This connects to Vaez et al. (2020) findings, suggesting that tourists become anti-social when they only trust their digital maps and do not need to talk, or even look at, the people they meet.

This way, tourists’ projects are goal-oriented, in the sense that the selection of stations and paths between them are optimized to reach the goal in the shortest time using the least resources possible. Phygital projects are thus carried out in physical space but are influenced by the perception of digital space through smartphones. As an additional consequence of smartphone use, the projects are rarely fixed and can change at any time, as in every time the tourists find new information (cf. Mieli, 2023).

The logic of optimization and efficiency is generally considered foreign to the idea of vacation, as mentioned by Olof (10). Reaching the destination effectively and efficiently might not be the primary goal when travelling for leisure, indeed getting lost or 'driving around' might be part of the experience. However, when using the phone to create mobility projects, the logic and the aim for optimising efficiency spills over from everyday life into the tourist experience, not only activities and functions as noted by Mieli (2022a) and Wang et al. (2016). This spillover effect may contribute to reduce the sense of magic, escape, and liminality associated with being a tourist (Benckendorff et al., 2019; Conti & Heldt Cassel, 2020; Pearce, 2011).

Bundles

The empirical data show that time-spatial bundles are joined both inside and outside tourists' vacation prisms. The analysis of bundles is organized around two sub-themes: digital bundles outside the physical vacation prism, and digital unbundling inside of it. Direct quotes from participants are summarized in Table 3 and referred to in the text.

Digital bundles outside the physical vacation prism. Tourists bundle with people in other places through digital connections. They stay connected with friends and family in other places, including back home. This connection with home, paradoxically, becomes part of the vacation, as Liisa (13) states. They also perform work-related activities like digital meetings or answering emails that are easy to conduct on-site, and that become part of the vacation. Connections with everyday life are also kept through tasks, such as banking, apartment hunting, and paying rents. In fact, keeping connections to everyday life may be a way to feel more relaxed, like in the case of Zoe (14) who did not have any desire to disconnect because she worried that she would not be able to relax unless she kept looking for an apartment during her holiday.

Previous literature has focused on the social aspects of co-presence, showing that tourists feel both near and far between home and destination. This is blurring the distinctions between home and away (de Souza e Silva, 2006; Pearce, 2011; White & White, 2007). Time geography is useful in calling attention to the places where things happen. Smartphones do not only allow people to connect with other people over distance but also allow them to access tools and to conduct activities that were previously linked to specific places, such as work or bank offices. Everyday functions and activities do not merely spill over into the tourist experience but have become integral parts of it. However, although this reduces the liminality of the experience, tourists are not necessarily bothered by it, as quotes in Table 3 show.

The idea that social obligations can be put aside during vacation (Zhang & Zhang, 2022) is absent from these tourists' intentions. The ability to work and continue everyday activities on vacation enables travel even when social and professional obligations keep tourists tied to everyday life. This, as Tan and Chen (2021) noted, makes otherwise impossible trips possible. Simultaneously, this challenges the idea that the tourist experience is an escape from everyday life.

Table 3. Participants’ quotes on bundles.

Theme	Sub-theme	Quotes
Bundles	Digital bundles outside the physical vacation prism	1. Ines ‘around 5–6 I know it’s time... It’s starting to be morning back home so I can call my family, or I can check WhatsApp’. 2. Liisa ‘[Being on vacation] of course it’s also sending pictures to your parents and WhatsApp and it’s like uh we’re here look at the pool’ 3. Zoe ‘I was looking for a new place to live so I was still very much connected [...] I was worried that apartment hunting would be a big damper on the holiday, and I wouldn’t be able to fully relax’. 4. Anastasia ‘I took care of all the communication (...) so a few times it was like – should I be here or should I be there?’ 5. Hazel ‘I’m not limited to where I am. Even though I’m away from home, I still get that contact with people back in Singapore, my friends I can’t see’
	Digitally unbundling within the vacation prism	6. Gabriela ‘I was actually kind of happy that I had something else to do as well other than just being a tourist. [...] Also it was nice to have a break from my co-traveller’ 7. Nuray ‘Sometimes it happened that I was too tired after touring a lot and mostly at dinner time everybody is like not talking to each other but just on their phones, that can happen’ 8. Ines ‘I got that complaint from my boyfriend like stop checking your emails when you’re outside of work, nobody is expecting you to reply’

Digital unbundling inside the vacation prism. Data show that bundles are affected by smartphone use inside the vacation prism, sometimes resulting in bundles becoming looser. Tourists can separate from their travel partner for a while, do different activities, and then re-connect later through online communication which allows micro-coordination (cf. Lamsfus et al., 2015). Furthermore, tourists can entirely disconnect from their travel partners without even leaving the physical bundle by immersing themselves in the digital context of the phone. People engage in tourism to strengthen social bonds, and tourism is a social practice that brings people closer together (Haldrup & Larsen, 2009). And yet, people take time off from their travel partners too, as shown by several of the quotes in Table 3, for example by Gabriela (17), and Nuray (18).

The process of unbundling can cause an undesired lack of connection to travel partners because of distractions caused by the phone, like Ines’ boyfriend who complained about her checking emails during the trip (Ines 23). While previous research has explored the kind of co-presence that can occur with people who are not in the travel party (cf. Lamsfus et al., 2015; Sheller & Urry, 2006), less attention has been given to what happens to the bundles that exist inside the vacation prism. Data show that unbundling phenomena may actually take place also in case tourists travel through phygital space. Smartphones offer a good way for unbundling from each other.

Concluding discussion: proposing a phygital time geography

This paper asks what happens to tourists’ time-spatial behaviour when the space they inhabit is phygital. This question is particularly important for tourism geographers because mobility in time and space is innate in the concept of tourism. While scholars have been theorizing how time and space are de-coupled, and how information and communication technologies challenge traditional concepts of time and space (Ash et al., 2018; Couclelis, 2009; de Souza e Silva, 2006; Frith, 2012; Thulin et al., 2020; Thulin & Vilhelmson, 2019), the notion that tourists move in space for a limited time remains valid.

Time and space are still fundamental for the definition of tourism, and they delineate the boundaries of the tourist experience. The *vacation prism* is still limited by time and by spatial coordinates. However, such delimitations are becoming more elastic and ontologically hybrid. Through the analysis of the empirical material, we have shown how physical and digital time-spaces are enmeshed and overlaid in the phygital tourist experience. In particular, we have used key concepts of time geography to explore such processes. A new model of phygital time geography is proposed (Figure 2). Here, we show that bundles can be both physical and phygital.

The time-geographic concepts of *path*, *stations*, *projects*, and *bundles* are all influenced by smartphone use. Orienting oneself in space is not a mere physical activity. Tourists rely on digital tools to position themselves in physical space. If these tools do not function correctly, tourists' ability to orient themselves in space is diminished and often doubted. A place is not 'learnt' in the same way with physical as with digital maps. While the former is static, unchanging, and gives an overview of the place, the latter is dynamic, personalized, and moves with the user, thus completely altering tourists' perceptions of space—and of people in it.

Through orientation and directions provided by digital tools, phygital paths are created. That is, these paths do not exist exclusively in physical or digital space, respectively, but in a physical space that is overlaid with digital information. The same logic applies to the various stations—destinations, sites, or attractions—between which tourists move. Their existence is not entirely physical, as they are mediated by digital information. Tourists may doubt whether a seemingly interesting place in front of them is indeed worth visiting when it is not marked as an attraction on the digital map. Projects of mobility within the vacation prism, then, also assume a new ontological connotation and become phygital themselves. All of this suggests that fewer tourists have a sense of where they are, while simultaneously trusting the

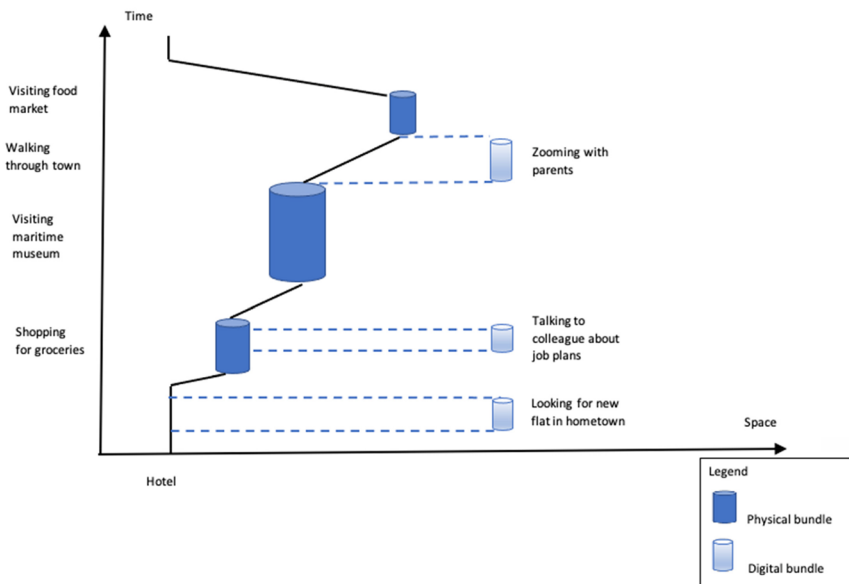


Figure 2. Phygital bundles in tourism space. Inspired by Thulin et al. (2020, p. 174).

device to let them know their position. We propose that this can be called a *sense of space*, meaning the (in)ability to locate oneself, and to be acquainted with one's surrounding. It resembles Massey's (2008) concept of sense of place but is allocated to wider spatial areas, and therein the relative position within and between places.

Another consequence is related to the liminal position of tourism. In time geography of everyday life, projects are goal oriented, and it is assumed that people try to achieve efficiency in their movements. This logic of optimization does not usually apply to tourism, where liminality, magic, and serendipity play a major part in creating a memorable experience (cf. Mieli, 2023; Conti & Heldt Cassel, 2020). However, by bringing everyday digital tools into the vacation space, tourists also bring the logic that accompanies them. Goal oriented efficiency and optimization spill over into the vacation through the digital devices that allow—and therefore afford—people to calculate optimal routes, reduce detours, avoid wandering about and getting lost. The project is phygital in the sense that it is performed in physical time space but shaped by digital information.

As observed by previous research, information and communication technologies allow for co-presence in hybrid spaces. Bundles inside and outside the physical vacation prism are shaped and influenced by the use of digital technology. This is despite the fact that tourist experiences are often pursued to achieve a sense of escape (Benckendorff et al., 2019; Pearce, 2011), and to strengthen social relations (Haldrup & Larsen, 2009). On one hand, tourists maintain digital bundles outside the vacation prism while still being physically present at the destination. On the other hand, a phenomenon of phygital unbundling takes place within the travel party, where individuals can momentarily dissolve the bundle through digital communication to other places.

This article has aimed to conceptualize a phygital time geography, adapted to technologically mediated tourists. It shows how tourist behaviour has changed due to smartphone use, following their influence on the perception of time-space. While showing that time geography is still a fertile perspective for understanding time-spatial behaviour, the paper has explored how tourists' ability to be in different places has changed, as with smartphones, tourists can not only travel between places, but they can digitally jump between them in a fraction of a second. Therefore, the present paper highlights the renewed relevance of time geography to address the question of the spatiotemporal dimensions of the tourist experience. Through the empirical examples, the paper shows how the geography of the tourist experience is mediated by technology in ways that challenge principles of time geography, as we state that time and space are no longer two inseparable dimensions (principle 1), and the inseparability of individuals can be questioned, when they are phygitaly active (principle 4). Future research on a digital time geography should take this into consideration.

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No potential conflict of interest was reported by the author(s).

Notes on contributors

Micol Mieli, PhD is a postdoctoral researcher at the Department of Communication and Arts at Roskilde University. After a BA in law, she later pursued an MSc and a PhD in service studies at

Lund University in Sweden. Micol's research has dealt with consumer behavior and tourism, particularly concerning technology, information search, and spatiotemporal behavior, adopting an interdisciplinary approach to explore the relationship between people and technology and the effects of digitalization on society. In her research, Micol has written about tourist information behaviour and tourists' relationship with mobile technologies, in particular the smartphone, drawing from tourism, human geography, and philosophy of technology. Micol also has a keen interest in epistemology and methodology, striving to develop innovative methodologies in her research.

Malin Zillinger, PhD is Associate professor at the Department of Service Studies, Lund University and researcher at the Department of Tourism Studies and Geography at Mid Sweden University. She has a PhD in Human Geography from Umeå University. She is interested in tourism geography, mobility and the intersection between tourism and everyday life. In her research projects, she studies people's mobility in time and space, and how digitalisation has affected the intersection of tourism, work, and leisure. Her study of mobility includes a special interest on method development.

Jan-Henrik Nilsson, PhD is an associate professor in human geography at the Department of Service Studies, Lund University, Sweden. His research interest lies mainly in the field of tourism geography, in recent years with an urban focus. His publications cover areas like urban tourism and destination development, sustainable urban tourism, ecosystem services, tourism and transport, aviation policy, food tourism, and the history of tourism and hospitality. Regionally, his research is mainly focused on the Nordic and Baltic Sea Regions.

ORCID

Micol Mieli  <http://orcid.org/0000-0002-1916-5548>

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